

Algebra 1  
Unit 6  
Lesson 1 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Jackson made a mistake when solving the inequality  $\frac{-8x+20}{4} - 10 \geq 7$ .  
For questions 1- 4, use Jackson's work below.

|                         |                                  |
|-------------------------|----------------------------------|
| <b>Given Inequality</b> | $\frac{-8x + 20}{4} - 10 \geq 7$ |
| <b>Step 1</b>           | $-8x + 20 - 10 \geq 28$          |
| <b>Step 2</b>           | $-8x + 10 \geq 28$               |
| <b>Step 3</b>           | $-8x \geq 18$                    |
| <b>Step 4</b>           | $x \leq -\frac{18}{8}$           |

1. Identify the step in which Jackson made an error. Write a note to Jackson explaining why that step was incorrect.

The error was made in Step 1. Before you can multiply by 4, the  $-10$  must be added to both sides.

2. Solve the inequality  $\frac{-8x+20}{4} - 10 \geq 7$ .

$$\frac{-8x+20}{4} \geq 17$$

$$4 \cdot \frac{-8x+20}{4} \geq 17 \cdot 4$$

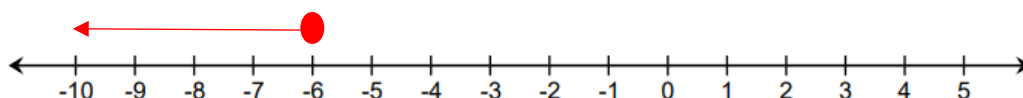
$$\begin{array}{r} -8x + 20 \geq 68 \\ -20 \quad -20 \end{array}$$

$$\frac{-8x}{-8} \geq \frac{48}{-8} \quad x \leq -6$$

3. Write the solution in set-builder notation.

$$\{x \in \mathbb{R} : x \leq -6\}$$

4. Graph the solution set.



Use the inequality  $0.05x + 1.13 > -0.24x + 14.18$  to complete questions 5-7.

5. Solve the inequality. 
$$\begin{array}{rcl} 0.05x + 1.13 & > & -0.24x + 14.18 \\ +0.24x & & + 0.24x \end{array}$$

$$\begin{array}{rcl} 0.29x + 1.13 & > & 14.18 \\ -1.13 & - & 1.13 \end{array}$$

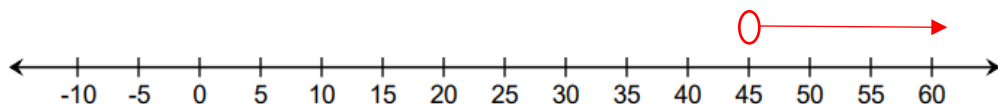
$$\frac{0.29x}{0.29} > \frac{13.05}{0.29}$$

$$x > 45$$

6. Write the solution in set-builder notation.

$$\{x \in \mathbb{R} : x > 45\}$$

7. Graph the solution set.



Use the inequality  $-\frac{3}{2}y + \frac{2}{3} \leq -\frac{1}{3}(y + 12)$  to complete questions 8-10.

8. Solve the inequality. 
$$\begin{array}{rcl} -\frac{3}{2}y + \frac{2}{3} & \leq & -\frac{1}{3}y - 4 \\ +\frac{1}{3}y & & +\frac{1}{3}y \end{array}$$

$$\begin{array}{rcl} -\frac{7}{6}y + \frac{2}{3} & \leq & -4 \\ -\frac{2}{3} & - & -\frac{2}{3} \end{array}$$

$$-\frac{6}{7} \cdot -\frac{7}{6}y \leq -\frac{14}{3} \cdot -\frac{6}{7}$$

$$y \geq 4$$

9. Write the solution in set-builder notation.

$$\{y \in \mathbb{R} : y \geq 4\}$$

10. Graph the solution set.

